The palindrome in the continued fraction of $\sqrt{n}$

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1 $\sqrt{2}$

Let $x = \sqrt{2}$. Then $x^2 - 1 = (x-1)(x+1) = 1$, so:

$$x \;=\; 1 + \frac{1}{1+x}.$$

Feed it into itself. Replacing the x in the denominator,

$$\sqrt{2} \;=\; 1 + \frac{1}{2 + \dfrac{1}{1+x}} \;=\; 1 + \frac{1}{2 + \dfrac{1}{2 + \dfrac{1}{1+x}}} \;=\; \cdots \;=\; 1 + \frac{1}{2 + \dfrac{1}{2 + \dfrac{1}{2 + \ddots}}}.$$

This is called a continued fraction.

2 $\sqrt{7}$

Let $x = \sqrt{7}$. Then $(x-2)(x+2) = x^2 - 4 = 3$, so $x - 2 = 3/(x+2)$, and:

$$x \;=\; 2 + \frac{3}{x+2}.$$

Feed it into itself. Adding 2 to both sides gives $x + 2 = 4 + 3/(x+2)$, so replacing the $x+2$ in the denominator,

$$\sqrt{7} \;=\; 2 + \frac{3}{4 + \dfrac{3}{x+2}} \;=\; 2 + \frac{3}{4 + \dfrac{3}{4 + \dfrac{3}{x+2}}} \;=\; \cdots \;=\; 2 + \frac{3}{4 + \dfrac{3}{4 + \dfrac{3}{4 + \ddots}}}.$$

This is nice, but it's not quite what we want. We want all the numerators to be 1. This can be obtained with some work.

Divide the top and bottom of $3/(x+2)$ by 3:

$$\sqrt{7} \;=\; 2 + \frac{1}{(x+2)/3}.$$

We need an expression for $(x+2)/3$, so subtract 1 to get $(x-1)/3$, and use $(x-1)(x+1)=6$ to get:

$$\frac{3}{x-1} \;=\; \frac{3(x+1)}{6} \;=\; \frac{x+1}{2}, \qquad \text{so} \qquad \sqrt{7} \;=\; 2+\cfrac{1}{1+\cfrac{1}{(x+1)/2}}.$$

Now we need an expression for $(x+1)/2$, so subtract 1 again, and use $(x-1)(x+1)=6$ again to get:

$$\frac{2}{x-1} \;=\; \frac{2(x+1)}{6} \;=\; \frac{x+1}{3}, \qquad \text{so} \qquad \sqrt{7} \;=\; 2+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{(x+1)/3}}}.$$

Now we need an expression for $(x+1)/3$, so subtract 1 again, and use $(x-2)(x+2)=3$ to get:

$$\frac{3}{x-2} \;=\; \frac{3(x+2)}{3} \;=\; x+2, \qquad \text{so} \qquad \sqrt{7} \;=\; 2+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{x+2}}}}.$$

Now we need an expression for $x+2$, so subtract 4, and use $(x-2)(x+2)=3$ again to get:

$$\frac{1}{x-2} \;=\; \frac{x+2}{3}, \qquad \text{so} \qquad \sqrt{7} \;=\; 2+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{4+\cfrac{1}{(x+2)/3}}}}}.$$

We are back where we started. We already have an expression for $(x+2)/3$, so:

$$\sqrt{7} \;=\; 2+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{4+\cfrac{1}{1+\ddots}}}}}.$$

3 The two theorems

The tower

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \ddots}}}$$

is written $[\,a_0;\, a_1, a_2, a_3, \ldots\,]$.

Theorem 1 (Lagrange)**.** *Let n be a positive integer that is not a perfect square. Then*

$$\sqrt{n} \;=\; [\,a_0;\; \overline{a_1, a_2, \ldots, a_L}\,]$$

for some $L \geq 1$.

The least such L is the *period*.

Theorem 2 (Legendre)**.** *With n and L as above and $a_0 = \lfloor \sqrt{n} \rfloor$,*

$$\sqrt{n} \;=\; [\,a_0;\; \overline{a_1, a_2, \ldots, a_{L-1},\; 2a_0}\,],$$

and

$$a_i \;=\; a_{L-i} \qquad (1 \leq i \leq L-1).$$

4 The field $\mathbb{Q}(\sqrt{n})$

Fix for the rest of the paper a positive integer n that is not a perfect square. Everything that follows happens inside a field just a little larger than the rationals.

Lemma 3. *$\sqrt{n}$ is irrational.*

Proof. Suppose $\sqrt{n} = p/q$ with p, q integers, $q > 0$ and $\gcd(p, q) = 1$. Then $p^2 = nq^2$, so q^2 divides p^2. But $\gcd(p, q) = 1$ gives $\gcd(p^2, q^2) = 1$, so $q^2 = 1$ and $n = p^2$ is a perfect square. □

Write

$$\mathbb{Q}(\sqrt{n}) \;=\; \left\{\, p + q\sqrt{n} \;:\; p, q \in \mathbb{Q} \,\right\} \;\subset\; \mathbb{R}.$$

This set is closed under addition and subtraction, and also under multiplication, because $\sqrt{n} \cdot \sqrt{n} = n$ stays inside:

$$(p + q\sqrt{n})(p' + q'\sqrt{n}) \;=\; (pp' + nqq') \;+\; (pq' + p'q)\sqrt{n}.$$

Division is the only thing needing an argument, and it is where Lemma 3 earns its keep.

Lemma 4. *Every element of $\mathbb{Q}(\sqrt{n})$ is $p + q\sqrt{n}$ for exactly one pair $(p, q) \in \mathbb{Q} \times \mathbb{Q}$, and every nonzero element has an inverse in $\mathbb{Q}(\sqrt{n})$. So $\mathbb{Q}(\sqrt{n})$ is a field.*

Proof. If $p + q\sqrt{n} = p' + q'\sqrt{n}$ with $q \neq q'$, then $\sqrt{n} = (p - p')/(q' - q)$ is rational, contradicting Lemma 3. So $q = q'$, and then $p = p'$.

Now let $\alpha = p + q\sqrt{n} \neq 0$. First, $p^2 - nq^2 \neq 0$: if $q = 0$ it is p^2 and $p \neq 0$; if $q \neq 0$ then $p^2 = nq^2$ would make $\sqrt{n} = \pm p/q$ rational. Multiplying numerator and denominator by $p - q\sqrt{n}$,

$$\frac{1}{p + q\sqrt{n}} \;=\; \frac{p - q\sqrt{n}}{p^2 - nq^2},$$

which is again of the required shape. □

The trick in that last line is the one every schoolchild learns as *rationalising the denominator*. Its two ingredients deserve names.

Definition 5. The *conjugate* and the *norm* of $\alpha = p + q\sqrt{n}$ are

$$\bar{\alpha} \;=\; p - q\sqrt{n}, \qquad N(\alpha) \;=\; \alpha\,\bar{\alpha} \;=\; p^2 - nq^2 \;\in\; \mathbb{Q}.$$

By the uniqueness half of Lemma 4 the conjugate is well defined. In analogy with the complex case it is a field automorphism: it clearly respects sums, and for products

$$\bar{\alpha}\,\bar{\beta} \;=\; (p - q\sqrt{n})(p' - q'\sqrt{n}) \;=\; (pp' + nqq') - (pq' + p'q)\sqrt{n} \;=\; \overline{\alpha\beta}.$$

It is an involution, $\bar{\bar{\alpha}} = \alpha$, and it fixes exactly the rationals, since $q = -q$ forces $q = 0$. Hence the norm is multiplicative,

$$N(\alpha\beta) \;=\; \alpha\beta\,\overline{\alpha\beta} \;=\; (\alpha\bar{\alpha})(\beta\bar{\beta}) \;=\; N(\alpha)N(\beta),$$

and the inverse computed above is just $\alpha^{-1} = \bar{\alpha}/N(\alpha)$.

One warning, because the whole proof of Theorem 2 turns on it: conjugation has nothing to do with the ordering of $\mathbb{R}$. It is an automorphism of the field, not of the ordered field, and α and $\bar{\alpha}$ are two unrelated real numbers. The strategy will be to arrange that $\alpha_i > 1$ while $\bar{\alpha}_i$ lies in $(-1, 0)$, and then to read the algorithm off the conjugates.

5 The general algorithm

Now let us look at the general algorithm. Write $\lfloor \alpha \rfloor$ for the largest integer not exceeding α. Then starting with $\alpha_0 = x = \sqrt{n}$ we have

$$\alpha_0 \;=\; \lfloor \alpha_0 \rfloor + \varepsilon_1,$$

where $0 < \varepsilon_1 < 1$ because x is irrational (Lemma 3). Turn ε_1 over to get $\alpha_1 = 1/\varepsilon_1 > 1$, and repeat:

$$\alpha_1 \;=\; \lfloor \alpha_1 \rfloor + \varepsilon_2, \qquad \alpha_2 \;=\; \frac{1}{\varepsilon_2},$$

and so on for ever. Writing $a_i = \lfloor \alpha_i \rfloor$,

$$\alpha_i \;=\; a_i + \varepsilon_{i+1}, \qquad \alpha_{i+1} \;=\; \frac{1}{\varepsilon_{i+1}}.$$

For $x = \sqrt{7}$ these are the numbers we met:

$$\begin{aligned}
\alpha_0 &= x, \\
\alpha_1 &= \frac{x+2}{3}, \\
\alpha_2 &= \frac{x+1}{2}, \\
\alpha_3 &= \frac{x+1}{3}, \\
\alpha_4 &= x+2, \\
\alpha_5 &= \alpha_1.
\end{aligned}$$

The α_i are the *complete quotients* and the a_i the *partial quotients*.

Look again at the complete quotients of $\sqrt{7}$. Every one of them has the shape

$$\frac{P+\sqrt{7}}{Q}$$

with P and Q whole numbers:

i	0	1	2	3	4
P_i	0	2	1	1	2
Q_i	1	3	2	3	1

Lemma 6. *Put $P_0 = 0$ and $Q_0 = 1$, and define*

$$P_{i+1} \;=\; a_i Q_i - P_i, \qquad Q_{i+1} \;=\; \frac{n - P_{i+1}^2}{Q_i}.$$

Then every P_i and Q_i is a whole number, and

$$\alpha_i \;=\; \frac{P_i + \sqrt{n}}{Q_i}.$$

Proof. This is proof by induction. Subtracting a_i leaves

$$\varepsilon_{i+1} \;=\; \frac{P_i+\sqrt{n}}{Q_i} - a_i \;=\; \frac{\sqrt{n} - (a_i Q_i - P_i)}{Q_i} \;=\; \frac{\sqrt{n} - P_{i+1}}{Q_i}.$$

Now:

$$\alpha_{i+1} \;=\; \frac{1}{\varepsilon_{i+1}} \;=\; \frac{Q_i}{\sqrt{n} - P_{i+1}} \;=\; \frac{Q_i\,(P_{i+1} + \sqrt{n})}{n - P_{i+1}^2} \;=\; \frac{P_{i+1} + \sqrt{n}}{Q_{i+1}}.$$

By assumption Q_i divides $n - P_i^2$. We must show that it divides $n - P_{i+1}^2$:

$$\begin{aligned}
n - P_{i+1}^2 \;&=\; n - (a_i Q_i - P_i)^2 \\
&=\; (n - P_i^2) + Q_i\, a_i (2P_i - a_i Q_i),
\end{aligned}$$

which is divisible by Q_i. So Q_{i+1} is a whole number, and it divides $n - P_{i+1}^2$ because $Q_i Q_{i+1} = n - P_{i+1}^2$. □

6 Only finitely many P and Q

We now have an algorithm for calculating P and Q:

```
from math import isqrt

def pq(n, steps):
    a0 = isqrt(n)
    P, Q = 0, 1
    for _ in range(steps):
        a = (a0 + P) // Q
        P = a * Q - P
        Q = (n - P * P) // Q
        print(a, P, Q)
```

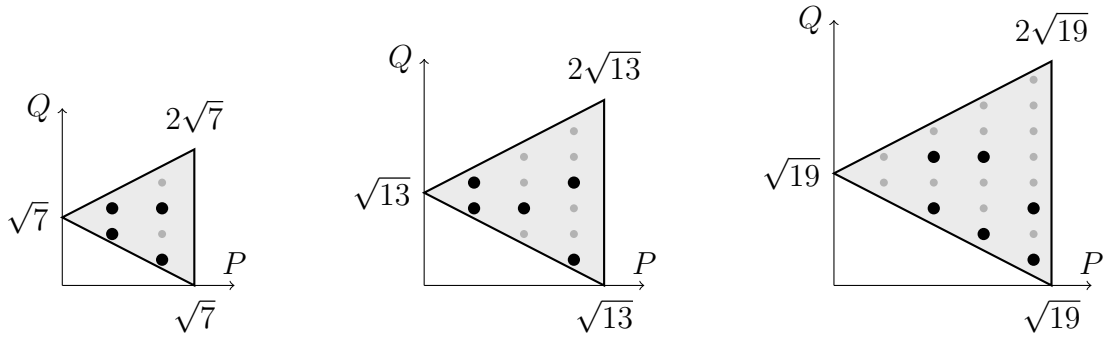
One period of it, for three values of n:

i	1	2	3	4
P_i	2	1	1	2
Q_i	3	2	3	1

i	1	2	3	4	5
P_i	3	1	2	1	3
Q_i	4	3	3	4	1

i	1	2	3	4	5	6
P_i	4	2	3	3	2	4
Q_i	3	5	2	5	3	1

Plot them:



$n = 7$ $n = 13$ $n = 19$

We can think of this as a dynamical system. What is surprising is that the values of (P, Q) always lie in the shaded triangle. Hence there are only finitely many of them.

Lemma 7. *For $i \geq 1$ the pair (P_i, Q_i) lies in the triangle:*

$$0 < P_i < \sqrt{n} \qquad \textit{and} \qquad \sqrt{n} - P_i \;<\; Q_i \;<\; \sqrt{n} + P_i.$$

Proof. Induction on i. The base case is trivial, so suppose

$$0 < P_i < \sqrt{n} \qquad \text{and} \qquad \sqrt{n} - P_i \;<\; Q_i \;<\; \sqrt{n} + P_i.$$

Two consequences, used throughout. The left half gives $Q_i > \sqrt{n} - P_i > 0$. The right half says $P_i + \sqrt{n} > Q_i$, that is $\alpha_i > 1$, so $a_i = \lfloor \alpha_i \rfloor \geq 1$.

Step 1: $Q_i < \sqrt{n} + P_{i+1}$. Rearrange the recurrence $P_{i+1} = a_i Q_i - P_i$ of Lemma 6:

$$a_i\, Q_i \;=\; P_i + P_{i+1}.$$

Since $a_i \geq 1$ and $Q_i > 0$ we may drop the a_i, and since $P_i < \sqrt{n}$ we may enlarge the right-hand side:

$$Q_i \;\leq\; a_i Q_i \;=\; P_i + P_{i+1} \;<\; \sqrt{n} + P_{i+1}.$$

Step 2: $0 < \sqrt{n} - P_{i+1} < Q_i$. Multiply $\alpha_i = a_i + \varepsilon_{i+1}$ by Q_i. By Lemma 6 the left side is $P_i + \sqrt{n}$, and by Step 1 the term $a_i Q_i$ is $P_i + P_{i+1}$, so

$$\varepsilon_{i+1}\, Q_i \;=\; (P_i + \sqrt{n}) - (P_i + P_{i+1}) \;=\; \sqrt{n} - P_{i+1}.$$

Now $0 < \varepsilon_{i+1} < 1$ and $Q_i > 0$ give the claim.

Step 3: $0 < P_{i+1} < \sqrt{n}$. Steps 1 and 2 together put Q_i strictly between two numbers:

$$\sqrt{n} - P_{i+1} \;<\; Q_i \;<\; \sqrt{n} + P_{i+1}.$$

If P_{i+1} were ≤ 0 the right number would not exceed the left, and no such Q_i could exist. So $P_{i+1} > 0$; and $P_{i+1} < \sqrt{n}$ by Step 2.

Step 4: the bounds for Q_{i+1}. By Lemma 6,

$$Q_i\, Q_{i+1} \;=\; n - P_{i+1}^2 \;=\; (\sqrt{n} - P_{i+1})(\sqrt{n} + P_{i+1}).$$

So Q_i and Q_{i+1} are a pair of factors of that product, and each bound on Q_i from Step 3 becomes the opposite bound on Q_{i+1}: from $Q_i < \sqrt{n} + P_{i+1}$ we get $Q_{i+1} > \sqrt{n} - P_{i+1}$, and from $Q_i > \sqrt{n} - P_{i+1}$ we get $Q_{i+1} < \sqrt{n} + P_{i+1}$. □

7 Möbius transformations

We already know that the fraction $1/2$ is the same as the fraction $2/4$, and more generally that

$$\frac{x}{y} \;=\; \frac{\lambda x}{\lambda y} \qquad (\lambda \neq 0).$$

In particular, if $xy' = x'y$ then taking $\lambda = y'/y$ we have

$$\lambda x \;=\; \frac{xy'}{y} \;=\; \frac{x'y}{y} \;=\; x', \qquad \lambda y \;=\; \frac{yy'}{y} \;=\; y',$$

so

$$\frac{x'}{y'} \;=\; \frac{\lambda x}{\lambda y} \;=\; \frac{x}{y}.$$

This allows us to define fractions as equivalence classes of pairs of integers:

$$\mathbb{Q} \;=\; \big\{(x,y) \in \mathbb{Z} \times \mathbb{Z} \;:\; y \neq 0\big\} / \sim, \qquad (x,y) \sim (x',y') \iff xy' = yx'.$$

Let us now add a *point at infinity*:

$$\widehat{\mathbb{Q}} \;=\; \mathbb{Q} \cup \{\infty\}.$$

A *point* is an equivalence class, written $[x : y]$. There are two kinds of class:

$$\begin{aligned} y \neq 0: &\quad [x:y] \;=\; [x/y : 1], && \text{an ordinary fraction;} \\ y = 0: &\quad [x:0] \;=\; [1:0], && \text{one new point, written } \infty. \end{aligned}$$

Definition 8. Let $A = \begin{pmatrix} r & s \\ u & v \end{pmatrix}$ with $\det A \neq 0$. Its *Möbius transformation* acts on $\widehat{\mathbb{Q}}$ by matrix multiplication:

$$A \cdot [x : y] \;=\; [\, rx + sy \;:\; ux + vy \,].$$

This is well defined on classes:

$$A \cdot [\lambda x : \lambda y] \;=\; [\, \lambda(rx + sy) \;:\; \lambda(ux + vy) \,] \;=\; [\, rx + sy \;:\; ux + vy \,] \;=\; A \cdot [x : y].$$

For example

$$J \;=\; \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad J \cdot [x : y] \;=\; [\, y : x \,],$$

is the map $z \mapsto 1/z$. It sends $0 = [0 : 1]$ to $[1 : 0] = \infty$, and ∞ back to 0.

More generally:

$$A \cdot [x : 1] \;=\; [\, rx + s \;:\; ux + v \,] \;=\; \left[\, \frac{rx + s}{ux + v} \;:\; 1 \,\right],$$

And ∞ is sent to the fraction r/u:

$$A \cdot \infty \;=\; A \cdot [1 : 0] \;=\; [\, r : u \,],$$

which is ∞ again when $u = 0$.

That last step of the first display divides by $ux + v$. At most one x has $ux + v = 0$, namely $x = -v/u$:

$$A \cdot \left[\, -\frac{v}{u} \;:\; 1 \,\right] \;=\; \left[\, \frac{su - rv}{u} \;:\; 0 \,\right] \;=\; \infty.$$

So ∞ just behaves like an ordinary point.

8 Conjugating the algorithm

Every complete quotient $\alpha_i = (P_i + \sqrt{n})/Q_i$ lives in $\mathbb{Q}(\sqrt{n})$, so we may conjugate it. Since conjugation is a field automorphism, applying it to $\alpha_i = a_i + 1/\alpha_{i+1}$ — in which the a_i are rational, hence fixed — gives

$$\bar{\alpha}_i \;=\; a_i + \frac{1}{\bar{\alpha}_{i+1}}.$$

For $i \geq 1$ Lemma 7 gives $0 < \sqrt{n} - P_i < Q_i$, so

$$\bar{\alpha}_i \;=\; \frac{P_i - \sqrt{n}}{Q_i} \;\in\; (-1, 0).$$

Now put

$$\beta_i \;=\; -\frac{1}{\bar{\alpha}_i} \;=\; \frac{Q_i}{\sqrt{n} - P_i} \;=\; \frac{Q_i\,(\sqrt{n} + P_i)}{n - P_i^2} \;=\; \frac{P_i + \sqrt{n}}{Q_{i-1}} \;>\; 1,$$

the last step by $Q_{i-1}Q_i = n - P_i^2$.

Lemma 9. *For $i \geq 1$,*

$$\beta_{i+1} \;=\; a_i + \frac{1}{\beta_i}, \qquad a_i \;=\; \lfloor \beta_{i+1} \rfloor.$$

Proof. $1/\bar{\alpha}_{i+1} = \bar{\alpha}_i - a_i$, so

$$\beta_{i+1} \;=\; a_i - \bar{\alpha}_i \;=\; a_i + \frac{1}{\beta_i},$$

and $\beta_i > 1$. □

9 Proof of Theorem 1

Proof. By Lemma 4 the representation $\alpha_i = P_i/Q_i + (1/Q_i)\sqrt{n}$ is unique, so α_i determines P_i and Q_i. By Lemma 7 there are finitely many pairs available for $i \geq 1$, so $\alpha_j = \alpha_k$ for some $1 \leq j < k$. Choose such a pair with k least, and among those j least.

Suppose $j \geq 2$. Then $P_j = P_k$ and $Q_j = Q_k$, hence $Q_{j-1} = (n - P_j^2)/Q_j = Q_{k-1}$ and $\beta_j = \beta_k$. By Lemma 9, $a_{j-1} = \lfloor \beta_j \rfloor = \lfloor \beta_k \rfloor = a_{k-1}$, and so

$$\alpha_{j-1} \;=\; a_{j-1} + \frac{1}{\alpha_j} \;=\; a_{k-1} + \frac{1}{\alpha_k} \;=\; \alpha_{k-1},$$

contradicting the minimality of k. So $j = 1$, and with $L = k - 1$,

$$\alpha_{L+1} \;=\; \alpha_1.$$

□

10 Proof of Theorem 2

Proof. $P_{L+1} = P_1$ and $Q_{L+1} = Q_1$, so

$$Q_L \;=\; \frac{n - P_{L+1}^2}{Q_{L+1}} \;=\; \frac{n - P_1^2}{Q_1} \;=\; Q_0 \;=\; 1,$$

and $P_1 = a_0 Q_0 - P_0 = a_0$. Now:

$$\beta_{L+1} \;=\; \frac{P_{L+1} + \sqrt{n}}{Q_L} \;=\; a_0 + \sqrt{n} \;=\; \beta_1, \qquad a_L \;=\; \lfloor \beta_{L+1} \rfloor \;=\; 2a_0.$$

Expand $\beta_1 = a_0 + \sqrt{n}$ in two ways. Its integer part is $2a_0$ and

$$\beta_1 - 2a_0 \;=\; \sqrt{n} - a_0 \;=\; \frac{1}{\alpha_1},$$

so $\beta_1 = [\, 2a_0;\, a_1, a_2, a_3, \ldots\,]$, which is

$$\beta_1 \;=\; [\, \overline{2a_0;\; a_1, a_2, \ldots, a_{L-1}}\,].$$

The other way is Lemma 9 run downwards from $\beta_{L+1} = \beta_1$:

$$\beta_1 \;=\; a_L + \cfrac{1}{a_{L-1} + \cfrac{1}{\ddots + \cfrac{1}{a_1 + \cfrac{1}{\beta_1}}}} \;=\; [\,\overline{2a_0;\; a_{L-1}, a_{L-2}, \ldots, a_1}\,].$$

Continued fraction expansions are unique, so

$$a_i \;=\; a_{L-i} \qquad (1 \le i \le L-1).$$

□

11 Checks and a consequence

n	block	palindromic part
7	$1,1,1,4$	$1,1,1$
13	$1,1,1,1,6$	$1,1,1,1$
19	$2,1,3,1,2,8$	$2,1,3,1,2$
31	$1,1,3,5,3,1,1,10$	$1,1,3,5,3,1,1$
61	$1,4,3,1,2,2,1,3,4,1,14$	$1,4,3,1,2,2,1,3,4,1$
94	$1,2,3,1,1,5,1,8,1,5,1,1,3,2,1,18$	$1,2,3,1,1,5,1,8,1,5,1,1,3,2,1$

The symmetry of B was checked directly for these, and the theorem itself for every non-square below 10^6.

One consequence is worth recording, since it is what the palindrome is usually wanted for. The identity $n = P_i^2 + Q_i Q_{i-1}$, which is the recurrence for Q rearranged, becomes a sum of two squares exactly when two consecutive Q agree. That happens at the mirror line of the palindrome, and the mirror falls *between* two terms precisely when $L-1$ is even, that is when L is odd. For a prime p the period is odd if and only if $p = 2$ or $p \equiv 1 \pmod 4$, and one recovers Fermat's theorem that such primes are sums of two squares, together with an algorithm for finding the two squares.